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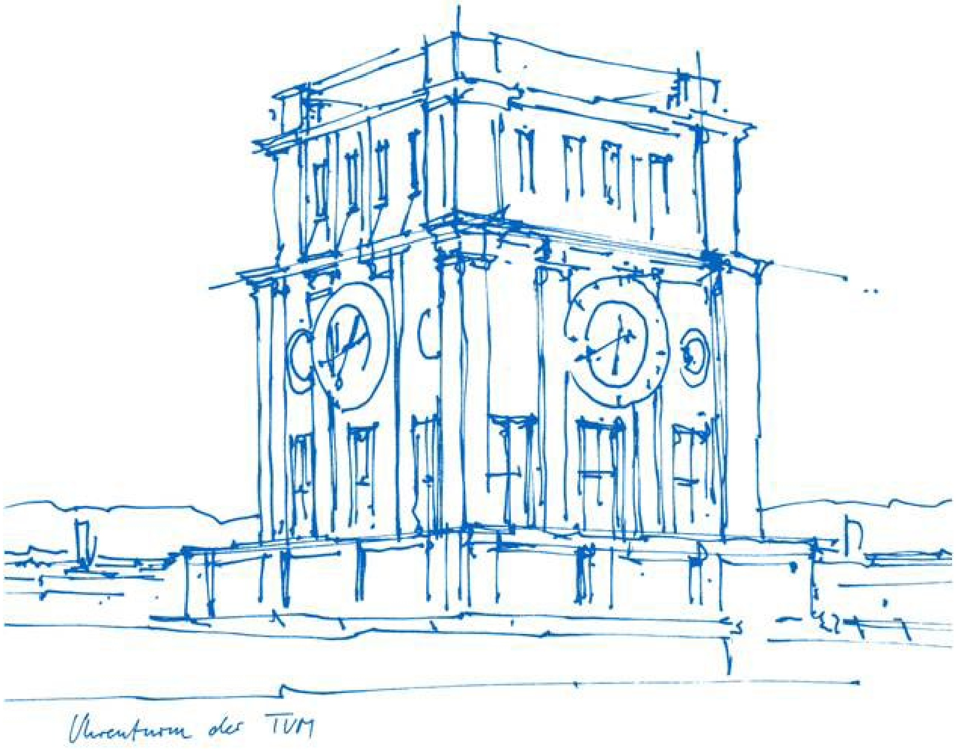
# Geo-Information

## E2 – Symbolization in a GIS




Winter Semester 2025 / 2026

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# Outline

- Quiz – L1
- Wrap-Up – E0&E1
- Introduction – E2
- Exercise Time – E2

# Quiz Session

- Visit tum.onlineted.de to access the quiz

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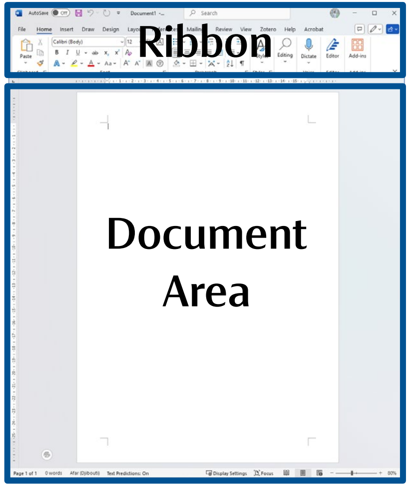
# Wrap-Up: User Interface (UI) of ArcGIS Pro

➢ **Draw a diagram showing the UI of ArcGIS Pro, including the following components:**

Ribbon, Tab, Group, Contents, Catalog, (Map) View, Attribute Table




An Example Diagram of Microsoft Word

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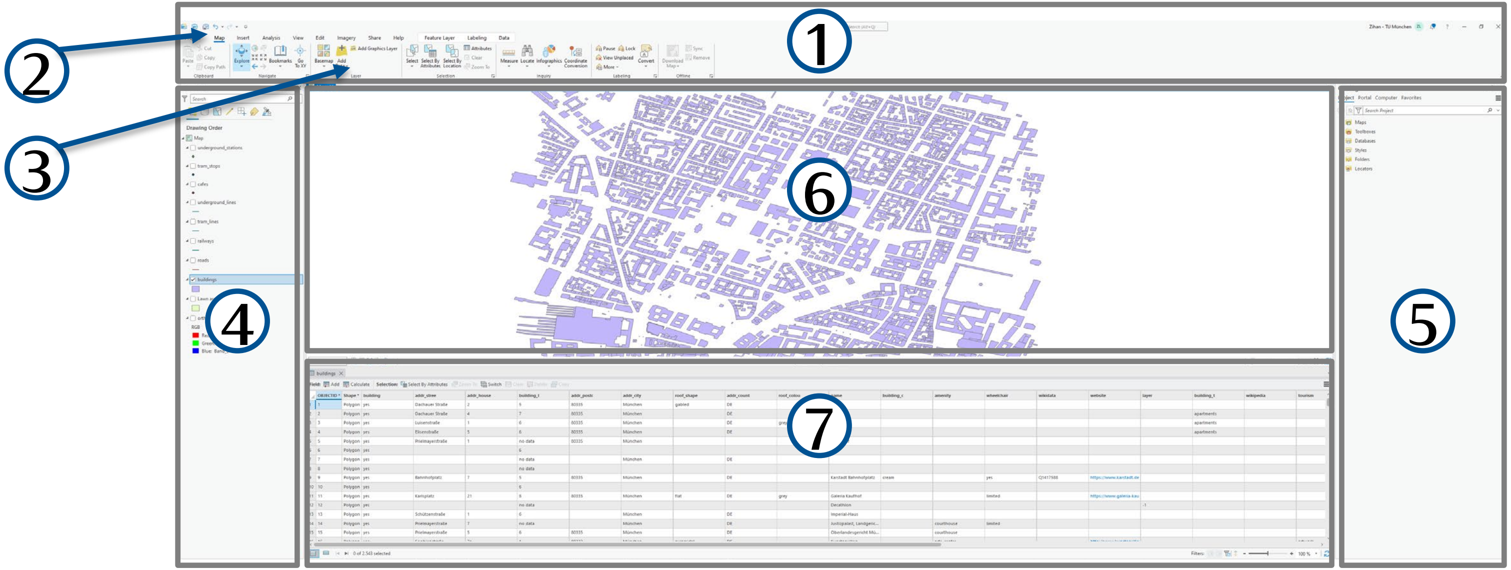
# Wrap-Up: User Interface (UI) of ArcGIS Pro

**Ribbon-Based UI**



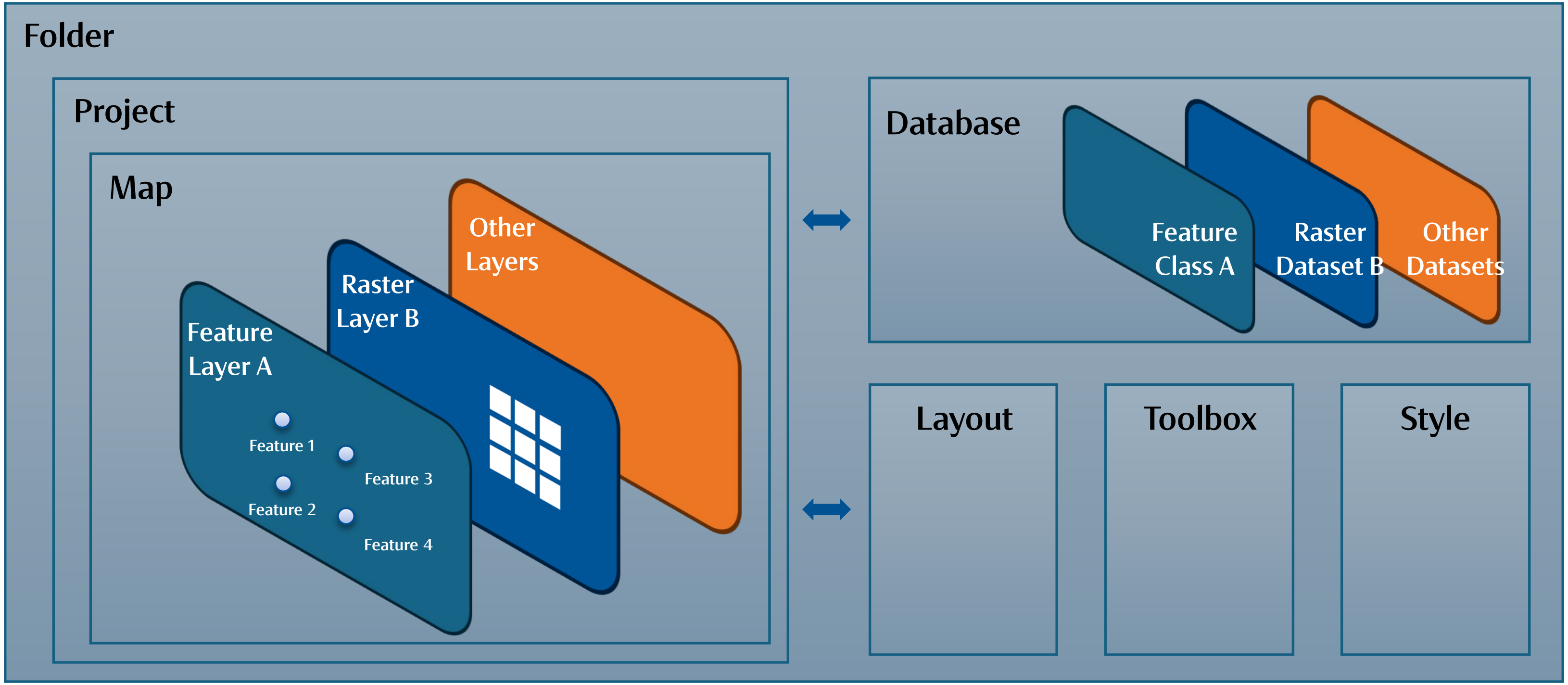

Ribbon – Tab – Group – Contents – Catalog – (Map) View – Attribute Table

# Wrap-Up: Core Concepts of ArcGIS Pro

| | | | | | |
|---|---|---|---|---|---|
| Folder | Project | Map | Database | Layer (Feature Layer, Raster Layer) | |
| Feature Class | Feature | Attribute Table | Field | Attribute | Geometry |

- ➢ Draw a Knowledge Graph to visualize the hierarchical relationships among these concepts.

# Wrap-Up: Core Concepts of ArcGIS Pro



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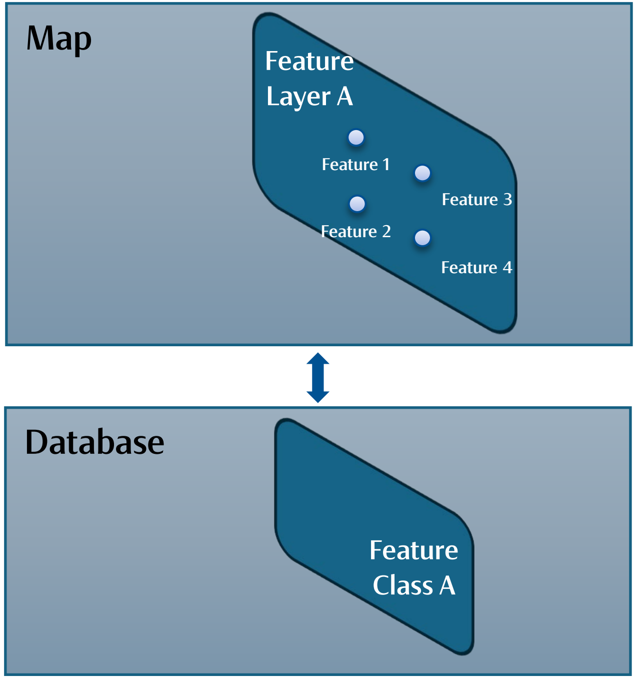
# Wrap-Up: Core Concepts of ArcGIS Pro




Attribute Table A

| | Field I | Field II | Other Fields | *Geometry (Shape)* |
|---|---|---|---|---|
| Feature 1 | Attribute 1-I | Attribute 1-II | ... | *Geometry 1* |
| Feature 2 | Attribute 2-I | Attribute 2-II | ... | *Geometry 2* |
| Feature 3 | ... | ... | ... | ... |
| Feature 4 | ... | ... | ... | ... |

**Attribute Table = Features × (Attributes + Geometry)**

# **Feature** in GIScience vs. in Data Science

| | Field I | Field II | Other Fields | *Geometry (Shape)* |
|---|---|---|---|---|
| Feature 1 | Attribute 1-I | Attribute 1-II | … | *Geometry 1* |
| Feature 2 | Attribute 2-I | Attribute 2-II | … | *Geometry 2* |
| Feature 3 | … | … | … | … |
| Feature 4 | … | … | … | … |

Attribute Table A (of Feature Class A)

| | Feature I | Feature II | Other Features |
|---|---|---|---|
| Observation / Record 1 | Variable 1-I | Variable 1-II | … |
| Observation / Record 2 | Variable 2-I | Variable 2-II | … |
| Observation / Record 3 | … | … | … |
| Observation / Record 4 | … | … | … |

Data Table A

**Attribute Table = Features × (Attributes + Geometry)**

**Data Table = Records × Features**

- The distinction lies mainly in **terminology** and the emphasis on **geometry**.

# Wrap-Up: E1

Selecting ⟵ **Feature** ⟶ Editing

- Interactive Feature Selection (Select by Shapes)
- Select by Attributes
- Select by Location

- Creating a new Feature Class
  - Point, Line, Polygon…
- Creating a new Feature
- Editing Fields and Attributes of Features
  - Adding new Fields
  - Renaming Fields
  - Editing Fields
  - Calculating Fields

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# Update on E1

## Basic Tasks

- [ ] **Creating a New Project** (1.2)
    - Connecting a Folder to the Project (1.2.1)
    - Adding a Database (1.2.2)
- [ ] **Adding a Layer**
    - Adding a Vector Layer (1.3)
    - Adding a Raster Layer (1.4)
- [ ] **Selecting Features**
    - Using a Pop-Up to Display a Feature's Attributes (1.5)
    - Selecting by Interactive Feature Selection (1.6)
    - Selecting by Location (1.7)
    - Selecting by Attributes (1.8)
- [ ] **Creating a New Feature Class** (1.10)
- [ ] **Editing New Features** (1.11)
    - Creating New Features (1.11.1)
    - Creating a New Template for Feature Creation (1.11.2)
- [ ] **Editing Fields and Attributes of Features** (1.9, 1.12)
    - Adding New Fields (1.12.1)
    - Renaming Fields (1.9.1)
    - Editing Fields (1.12.2)
    - Converting a String Field to a Numeric Field (1.9.2)
    - Calculating Fields (1.12.3)
    - Combining Editing Skills (1.12.4)

## Basic Tasks

- [ ] **Creating a New Project** (1.2)
    - Creating a New *Map* Project
    - Connecting a Folder to the Project (1.2.1)
        - Connect the folder where the geodatabase `Data Students.gdb` is located to the project.
    - Adding a Database (1.2.2)
        - Add the geodatabase `Data Students.gdb` to the project.
- [ ] **Adding a Layer**
    - Adding a Vector Layer (1.3)
        - Add `roads`, `railways` and `buildings` layers from the geodatabase to the map.
    - Adding a Raster Layer (1.4)
        - Add the raster image `orthophoto.tiff` from the geodatabase to the map.
- [ ] **Selecting Features**
    - Using a Pop-Up to Display a Feature's Attributes (1.5)
    - Selecting by Interactive Feature Selection (1.6)
        - Select all the TUM buildings using *Select by Polygon* or other shapes.
    - Selecting by Location (1.7)
        - Select all `cafés` around the TUM buildings within a `500-meter` radius.
    - Selecting by Attributes (1.8)
        - Select all tram lines (`fclass=tram`) and underground lines (`fclass=subway`) of the `railways` layer.
        - Copy the selected features into a separate layer, respectively named `tram_lines` and `underground_lines`.
- [ ] **Creating a New Feature Class** (1.10)
    - Create a new polygon feature class named `Lawn` in your project geodatabase.
    - Create two new point feature classes named `tram_stops` and `underground_stations`.
- [ ] **Editing New Features** (1.11)
    - Creating New Features (1.11.1)
        - In the `Lawn` layer, digitize the rectangular lawn around the Old Pinakothek as a new feature, using the orthophoto as reference.
        - In the `tram_stops` and `underground_stations` layers, digitize one point per stop or station using the MVG map (Page 18 in Lesson 1) as reference.
    - Creating a New Template for Feature Creation (1.11.2)
- [ ] **Editing Fields and Attributes of Features** (1.9, 1.12)
    - Renaming Fields (1.9.1)
        - In the `roads` layer, rename the field `highway` to `type` (alias: `type of road`) and highlight it.
        - Keep only the fields `OBJECTID`, `type`, `name`, `maxspeed`, and `oneway` visible.
    - Converting a String Field to a Numeric Field (1.9.2)
        - In the `roads` layer, convert the string field `maxspeed` into a numeric field `speed_num`.
    - Adding New Fields (1.12.1)
        - Add a new text field named `name` to both the tram_stops and underground_stations layers.
    - Editing Fields (1.12.2)
        - For both `tram_stops` and `underground_stations` layers, update the `name` field directly in the Attribute Table by typing correct names from the MVG plan.
    - Calculating Fields (1.12.3)
        - Select all TUM building polygons and calculate the field `amenity` in the buildings layer.
        - Use the *Calculate Field* tool, assign the value `TUM` to the selected records.
    - Combining Editing Skills (1.12.4)
        - In the buildings layer, replace empty values in `building_l` with the text `no data`.
        - Create a new numeric field `level_num` and calculate its values from `building_l`.

# Introduction: E2

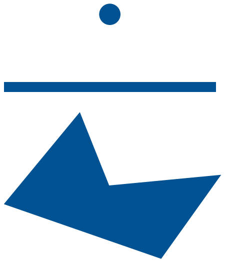
**Feature**

| | **Field I** | **Field II** | **Other Fields** | ***Geometry (Shape)*** |
|---|---|---|---|---|
| Feature 1 | Attribute 1-I | Attribute 1-II | ... | *Geometry 1* |
| Feature 2 | Attribute 2-I | Attribute 2-II | ... | *Geometry 2* |
| Feature 3 | ... | ... | ... | ... |
| Feature 4 | ... | ... | ... | ... |

Attribute Table A (of Feature Class A)

# Introduction: Feature Symbolization

**Attributes of Features** → **Visual Variables** → **Symbolization**



Point

Line

Polygon

München Annotation

Size
Shape
Color Hue
Color Value
Color Saturation
Orientation
Texture
…

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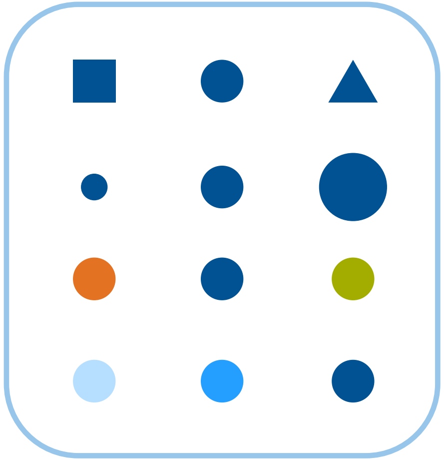
# Introduction: Visual Variables

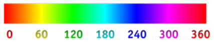
➢ How to choose a proper visual variable?

The relationship between a **symbol** and the **attribute** it represents be clear and easily interpretable.

And the choice primarily depends on whether the data are **qualitative** (categorial) or **quantitative** (numerical).



| Visual Variable | Feature Attribute |
|---|---|
| Shape | Qualitative |
| Size | Quantitative |
| Color Hue | Qualitative |
| Color Value | Quantitative |



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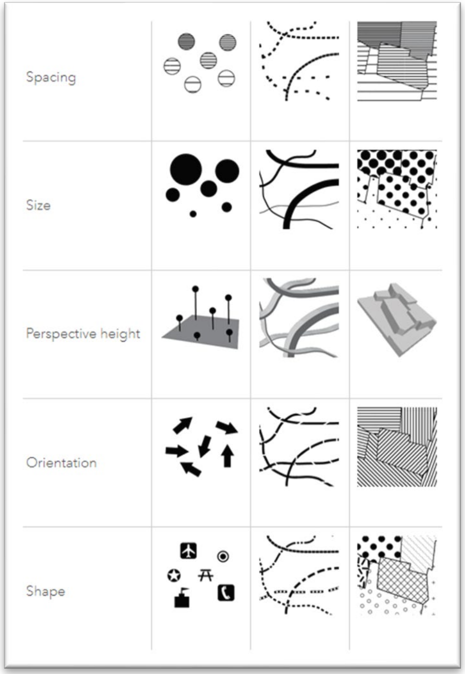
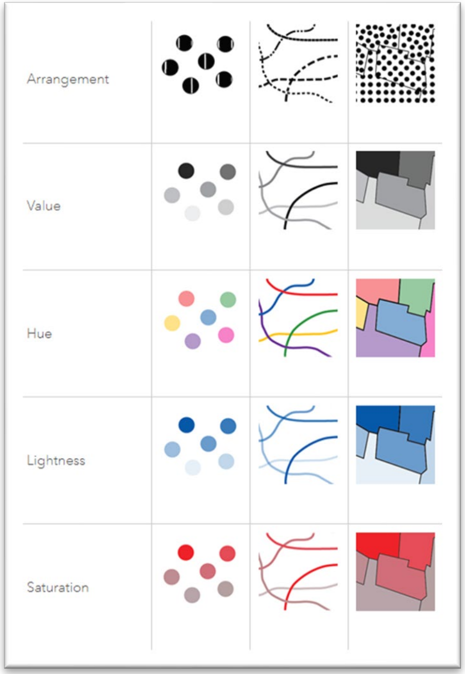
# Introduction: Visual Variables in ArcGIS Pro




Source: ArcGIS Pro

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# Introduction: Basic Tasks

- Uniform Symbolization
- Qualitative Symbolization
- Quantitative Symbolization
- Multi-Layer Symbolization
- Scale-Based Symbolization

➢ **Results of E1 is required to finish the tasks.**

(1.8, 1.9.2, 1.11.1, 1.12.3, 1.12.4)

- [ ] **Uniform Symbolization** (2.2)
  - Symbolizing Bus Stops with Single Symbol (2.2.1)
    - (Add `bus_stops` layer from the geodatabase.)
    - Symbolize bus stops using an appropriate point symbol from the gallery.
    - Adjust the color and size of the symbols if needed.
  - Symbolizing Tram Stops and Underground Stations with Single Symbol (2.2.2)
    - Symbolize tram stops and underground stations with external logo files `Tram_logo.svg` and `UBahn_logo.svg` from the `Symbols` folder.
    - Adjust the symbol size if needed.
- [ ] **Qualitative Symbolization** (2.3)
  - Symbolizing Museums and TUM Buildings with Unique Values (2.3.1)
    - Symbolize museums from the `buildings` layer in a unique color by its `tourism` field.
    - Symbolize TUM buildings from the `buildings` layer in a different unique color by its value `TUM` in the `amenity` field.
    - Symbolize all the other buildings in another different unique color.
- [ ] **Quantitative Symbolization** (2.4)
  - Symbolizing Buildings with Graduated Colors
    - Symbolize buildings with graduated colors by building heights, calculated with `level_num` and a standard story height of 3.5 m. (2.4.1)
    - Try different classification methods, classify and symbolize the building heights into reasonable classes. (2.4.2, 2.4.3)
    - Assign a neutral light grey color to the `no data` class. (2.4.3)
  - Symbolizing Roads with Proportional Symbols (2.4.4)
    - Symbolize roads with proportional line widths by the allowed maximum speed in the field `speed_num`.
    - Exclude the roads with null values and set an appropriate *Maximum Size* value.
- [ ] **Multi-Layer Symbolization** (2.5)
  - Symbolizing Roads with Cased Line Symbols
    - Symbolize all the car-traffic roads with cased line symbols by the field `type of road` (with the values `secondary`, `secondary_link`, `tertiary`, `residential`). (2.5.1)
    - Symbolize all the other roads with a thin line symbol in a light grey color. (2.5.1)
    - Join and merge the cased line symbols to show connectivity across individual road segments. (2.5.2)
- [ ] **Scale-Based Symbolization** (2.6)
  - Applying Scale-Based Symbol Classes to Roads (2.6.1)
    - Set all the car-traffic roads to be visible between 1:10,000 and larger.
    - Set all the other roads to be visible between 1:5,000 and larger.
  - Applying Scale-Based Symbol Sizing to Public Transport Points (2.6.2)
    - Set scale-based symbol sizing to all the bus stops, tram stops and underground stations (e.g., 50 pt at 1:1,000, ... 10 pt at 1:10,000, ... 1 pt at 1:100,000,000).
- [ ] **Symbolization with Transparency** (2.7)
  - Try the *Swipe* tool to compare `orthophoto` and `buildings` layer.
  - Adjust the *transparency* of the `buildings` layer so that the `orthophoto` shows through.
- [ ] **Combining Symbolization Skills** (2.8)
  - Symbolize the underground lines U1 - U6 using the styles shown below.

# Introduction: Advanced Tasks

- Symbolization in GIS
- Custom Color Ramp
- Bivariate Symbology
- Arcade Expressions
- Mixed Symbology
- From Symbology to Web Map Styles
- Visual Variables

- **[Symbology in GIS]** Compare and identify common symbology methods in ArcGIS Pro for points, lines, and polygons, and reflect on how different visual variables support qualitative and quantitative data.
- **[Custom Color Ramp]** Explore ColorBrewer to design a custom color ramp for building heights (or any other quantitative attribute), and apply it using *Graduated Colors* symbology.
- **[Bivariate Symbology]** Experiment with *Bivariate Symbology* in the buildings layer — for example, combining height and area to reveal multidimensional spatial patterns.
- **[Arcade Expressions]** Use *Arcade expressions* to create flexible, data-driven symbolization — for instance, by combining road type and speed limit into a single rule.
- **[Mixed Symbology]** Use *Vary Symbology by Attribute* to combine multiple visual variables within the same layer, such as size, color, rotation and transparency, and observe how this affects the map's readability.
- **[From Symbology to Map Styles]** Explore web map symbology using Maputnik, experiment with colors, line widths, and feature visibility across zoom levels, and reflect on how web map styles differ from symbology in ArcGIS Pro.
- **[Visual Variables]** Explore the dictionary of visual variables (e.g., here) and examine how each variable influences map perception.

# Introduction: Assignment Submission E2

- Submit two screenshots (.png or .jpg) in Moodle by **next Monday (03.11.2025)** showing the following results.
  - **Screenshot A:**
    - Results of 2.2: symbolized Public Transport Points (including Underground Stations, Tram Stops, and Bus Stops),
    - Results of 2.8: symbolized Underground Lines,
    - Symbolized Tram Lines with an appropriate style.
  - **Screenshot B:**
    - Results of 2.4.1 - 2.4.3: symbolized Buildings with graduated colors by building heights,
    - Results of 2.5: symbolized Roads with Cased Line Symbols.
- Please pay attention to the **drawing order** between layers, and only show the required results in each screenshot.